

Agent-Augmented Sequence Modeling for Explainable ICU Diagnosis Prediction

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ML Models for ICU Prediction

Machine learning models trained on ICU electronic health record time-series data can identify clinically meaningful patterns within the first hours of admission.

These models allow early prediction of diagnoses and outcomes, often outperforming traditional scoring systems like APACHE and SAPS.

RNNs and LSTM architectures demonstrate strong performance on multivariate clinical time-series tasks by capturing temporal dependencies in physiological data.

The Gap — & Our Approach

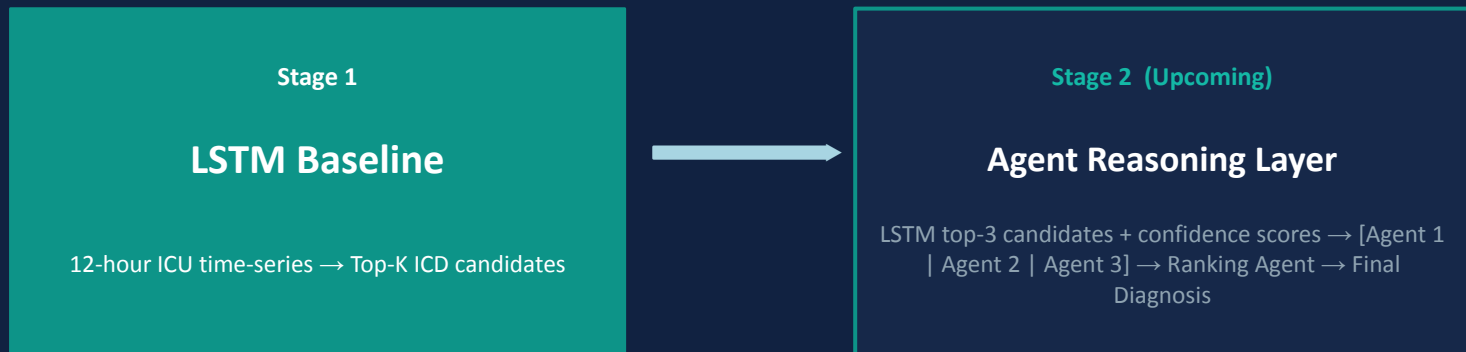
However, these models lack explicit clinical reasoning and may produce high-confidence errors when cases are ambiguous.

LLMs show promise in medical reasoning but remain unreliable on structured physiological data.

This motivates hybrid approaches. Multi-agent LLM systems can produce and evaluate multiple candidate explanations — but their use as an inference-time layer over trained clinical models remains largely unexplored.

Research Problem

LSTM models trained on ICU time-series produce high-confidence predictions but lack explicit clinical reasoning, making errors difficult to detect or correct. **Can the addition of a multi-agent LLM reasoning layer improve primary ICD-10 diagnosis prediction accuracy on held-out MIMIC-IV ICU codes, as measured by Top-1 accuracy and AUROC?**



Data Preprocessing: Cohort & Features

14,968

Patients

12 hr

Observation Window

502

Features

50

ICD-10 Classes

Cohort Inclusion Criteria

- ICU stay duration sufficient to yield at least 12 hourly observation windows
- At least one primary ICD-10-CM discharge diagnosis code present in the top-50 most frequent codes across the cohort
- Non-neonatal admissions; stays with incomplete or corrupt chartevents records excluded

Data Preprocessing: Feature Engineering

502-Dimensional Feature Vector

For each of the 251 clinical variables, two columns are produced:

Value Column (even index)

Observed measurement, or 0.0 if not recorded that hour

Mask Column (odd index)

Binary indicator: 1 = observed this hour, 0 = missing

Observation density: ~3.8% across all feature-hour pairs

Clinical Variables Included

chartevents

Vital signs, ventilator settings, nursing assessments (e.g., heart rate, SpO₂, GCS)

labevents

Laboratory results (creatinine, potassium, lactate, hemoglobin, INR, pH, pCO₂)

inputevents

Medication & fluid inputs (vasopressors, IV fluids, sedatives)

procedureevents

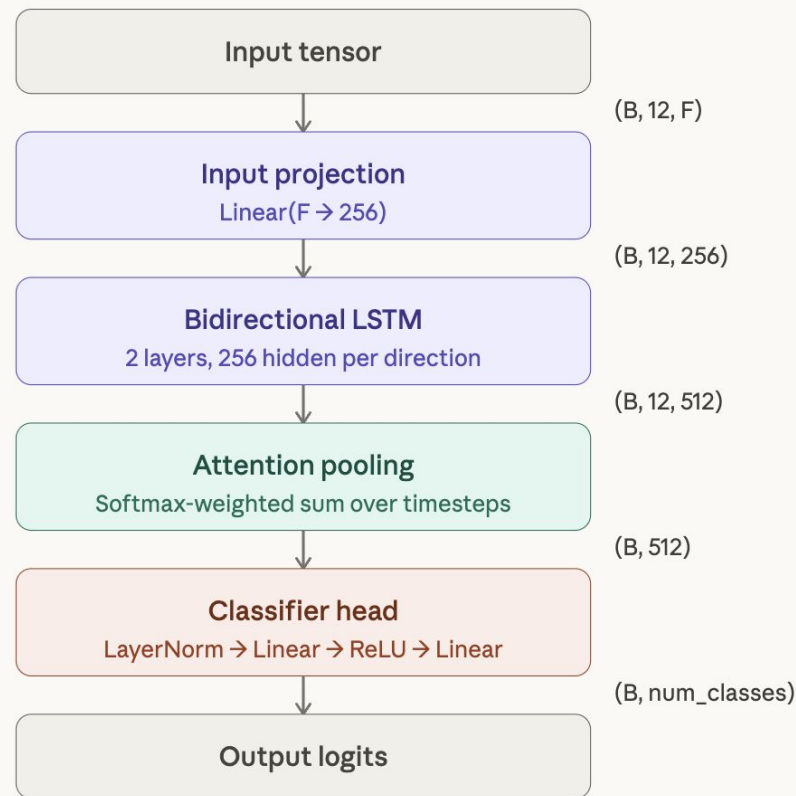
ICU procedures (intubation, dialysis, arterial line placement)

LSTM Architecture + Training

Split	Patients	Stays (ICU admissions)
Train	8,381 (70%)	10,485
Validation	1,795 (15%)	2,268
Test	1,797 (15%)	2,215
Total	11,973	14,968

Architecture Details

- Stacked **bidirectional** LSTM with fully-connected classification head
- Input:** 12 timesteps $\times$ 502 features
- Output:** probability distribution over 50 ICD-10 classes
- Loss:** cross-entropy with inverse-frequency class weights
- Early stopping:** monitors validation macro F1



Baseline Results · Test set: 2,215 stays

26.91%

Top-1 Accuracy

Fraction where top predicted ICD code matches true discharge code

72.87%

Top-5 Accuracy

Fraction where true code appears within the 5 highest-probability predictions

27.80% / 15.61%

F1 Weighted / Macro

Weighted: reflects common conditions ·
Macro: sensitive to rare class performance

Key Insight

The ~46 percentage-point gap between Top-1 (26.91%) and Top-5 (72.87%) accuracy reveals the critical opportunity: in many cases the correct diagnosis is in the LSTM's shortlist but ranked sub-optimally. This 'recoverable' ceiling is the core motivation for the LLM reasoning layer in Phase 2.

Challenges & Solutions

Identified Challenges

Class Imbalance

Up to 2,164:1 ratio in training. Macro F1 (13.5%) vs weighted F1 (25.3%) reveals severe rare-class underperformance.

Sparse Temporal Data

Given the set of 251 clinical features the dataset ends up being quite sparse.

ICD-10 as Proxy

Discharge codes are retrospective and billing-influenced; may not reflect the most prominent condition during admission.

Solutions

Class-balanced sampling (oversampling rare classes) vs. weighted loss alone

Mask-gated features to mitigate the sparsity.

Report weighted F1 alongside macro F1 to distinguish well-labeled common codes from noisy rare ones

Upcoming: Agent Reasoning Layer



- Optimize current baseline LSTM
- One specialist LLM agent per candidate diagnosis — each receives the structured clinical summary and its assigned ICD-10 label, and generates a clinical rationale for or against
- Ranking agent synthesizes all three rationales alongside LSTM confidence scores to select the final prediction
- No retraining — agent layer operates entirely at inference time on the frozen LSTM checkpoint

Thanks!

Feedback Ask:

We're debating what each agent should actually receive as input — right now it's the candidate diagnosis label plus the structured clinical summary. Should agents also see the LSTM confidence score for their assigned candidate, or would that bias the rationale toward confirming whatever the model already predicted?

References

- [1] Harutyunyan et al. Multitask learning and benchmarking with clinical time series data. *Scientific Data*, 6(1):96, June 2019.
- [2] Johnson et al. MIMIC-IV, a freely accessible electronic health record dataset. *Scientific Data*, 10:1, January 2023.
- [3] Nori et al. Capabilities of GPT-4 on Medical Challenge Problems. *arXiv:2303.13375*, April 2023.
- [6] Rao et al. Evaluating ChatGPT as an Adjunct for Radiologic Decision-Making. *medRxiv*, 2023.02.02.23285399, February 2023.
- [8] Banerjee et al. MedGellan: LLM-Generated Medical Guidance to Support Physicians. *arXiv:2507.04431*, September 2025.